OFFICIAL RULEBOOK & PROBLEM STATEMENTS

ZETHETA ALGORITHMS × IIT PATNA TRADING & INVESTMENT CLUB (TIC)

24-HOUR HACKATHON

1. EVENT OVERVIEW & STRUCTURE

Zetheta Algorithms Private Limited, knowledge partner to the IIT Patna TIC Inter-Collegiate Competition, sets four industry-grade financial problem statements. Teams must select exactly one track and deliver a end-to-end quantitative solution within 24 hours using publicly sourced Indian market data.

#	Track Name	Problem Statement Domain	Team Size
PS1	Equity Research & Consulting	Capital-markets infrastructure: one stock selection & IC memo	3–4 Members
PS2	Financial Risk & Fraud Analytics	Early-warning system (EWS) for corporate distress & manipulation	3–4 Members
PS3	Financial Forecasting	Quarter-end liquid fund redemption forecasting for AMC treasury	3–4 Members
PS4	AI-Powered Advisory	Compliant AI co-pilot for a SEBI-registered Investment Adviser (RIA)	3–4 Members

2. UNIVERSAL COMPETITION RULES

👥 Team Eligibility & Formation

Teams must consist of exactly 3 to 4 members. One problem statement must be formally declared at kick-off. Multi-team participation by a single individual is strictly prohibited.

🌐 Data Sourcing & Data Cut-Off

Only publicly available data may be used; every source must be explicitly cited. The cutoff is the last trading day prior to event start. Teams must acquire their own datasets.

🤖 Mandatory Tool Disclosure

AI models and LLMs are permitted and encouraged. However, teams must submit a declaration specifying exact AI tools used and their functional application.

⚙️ Reproducibility & Integrity

All figures, models, and tables must be 100% reproducible via submitted code/workbooks. Look-ahead bias or data leakage will result in immediate disqualification.

🔑 **Mandatory Repository Transfer Protocol:** All code and models must be hosted in a public GitHub repository. Prior to submission, ownership of the repository MUST be transferred to GitHub User ID: `ZethetaIntern`. Teams must communicate their original User ID and one registered email address to Zetheta for team mapping.

3. PROBLEM STATEMENT SPECIFICATIONS (PS1 & PS2)

PS1: Equity Research — Toll Collector of India's Capital Markets

TRACK 1

The Ask: Recommend which ONE of four listed capital-markets infrastructure companies a long-only fund should allocate ₹500 crore to, supported by a 12-month target price and a defensible bear case.

Candidate Company	Core Business	Key Revenue Driver
BSE Ltd	Stock Exchange	Transaction (esp. derivatives) volumes
CDSL	Depository	Demat accounts, transactions, KYC fees
CAMS	Mutual Fund RTA	AMC Assets Under Service (AUS)
KFin Technologies	RTA (MF, Issuer, Global)	AUM serviced, international expansion

Deliverables:

- **Industry Analysis:** Moats, pricing power, impact of SEBI F&O curbs and fee rationalization.
- **Valuation & FDD:** 3-statement DCF + relative valuation (P/E, EV/EBITDA); deep dive on cash conversion, capex, and related-party transactions.
- **Scenario Model:** Bull, base, and bear case (e.g., 30% drop in derivatives turnover, 20% equity drawdown).
- **IC Memo:** Concise 2-page investment committee memo with position sizing and 3 key exit triggers.

PS2: Risk & Fraud — Early-Warning System (EWS) for Default & Fraud

TRACK 2

The Ask: Build an Early-Warning System (EWS) that flags listed corporate borrowers heading for default or accounting manipulation at least 4 quarters in advance, optimized for a credit desk capacity of 5% (40/800 companies).

Deliverables & Evaluation Benchmarks:

- **Distress Model:** Trained on UCI Polish/Taiwanese Bankruptcy dataset using time-aware split.
- **Indian Case Test:** Panel of ≥10 defaulted Indian firms (IL&FS, DHFL, Jet, etc.) vs. 30 matched sector peers across 3 historical pre-event years.
- **Manipulation Screen:** Beneish M-Score, Piotroski F-Score, Altman Z-Score + soft signals (auditor resignations, promoter pledge ratios, rating changes).
- **Explainability & Policy:** SHAP value explanations per flagged firm; threshold tuning where a missed default costs **20x** a false positive.

4. PROBLEM STATEMENT SPECIFICATIONS (PS3 & PS4)

PS3: Financial Forecasting — Quarter-End AMC Redemption Forecast

TRACK 3

The Ask: Forecast monthly net flows for India's liquid and overnight fund categories with prediction intervals, and translate forecasts into an optimal cash-buffer policy to minimize idle cash drag vs. redemption penalties.

Core Methodology & Rules:

- **Forecast Engine:** 3-month forward point forecasts + 80% and 95% confidence intervals comparing classical models (SARIMAX/ETS) against ML/DL architectures.
- **Exogenous Drivers:** Advance tax dates (15th Mar/Jun/Sep/Dec), banking liquidity surplus/deficit, money market yields, and market returns.
- **Optimization Layer:** Buffer rule minimizing: `Cost = Idle Cash Drag + (3 × Shortfall Penalty)` backtested over 24 months (rolling-origin).
- **Fund Manager Dashboard:** 1-page operational interface for first-of-the-month positioning.

PS4: AI Advisory — Compliant Co-pilot for SEBI-Registered RIAs

TRACK 4

The Ask: Construct an AI co-pilot that automates client profiling, constructs goal-based mutual fund portfolios, answers SID/factsheet queries with page-level citations, and enforces strict SEBI suitability guardrails.

Persona Evaluation Matrix:

Target Persona	Age / Income Status	Primary Financial Goal
Fresh Graduate	23 yrs | ₹55k/mo take-home	Emergency fund creation → Long-term wealth
Young Couple	31 yrs | Dual income, home loan	Child education in 15 years
Small Business	44 yrs | Irregular cash flow	Structured retirement at age 58
Doctor (Patna)	38 yrs | High income, risk-averse	Retirement + Parent healthcare corpus
Retiree (Chennai)	68 yrs | ₹80L corpus + pension	Monthly income stream for 25 years
Aggressive Trader	29 yrs | F&O drawdown history	Requires suitability enforcement / risk flag

5. SUBMISSION GUIDELINES & ZECATHON PLATFORM

All registration, team creation, and project submissions must occur strictly on the **ZECATHON Platform**. Submissions via email, Drive links, or social media will be disqualified immediately.

Platform Field	Submission Requirement
Project Document (Required)	Report PDF (Max 8 pages; PS1: IC Memo + Appendices) OR Presentation Deck PDF (Max 10 slides). Must have selectable text (no scanned images).
Supporting GitHub URL	Public GitHub Repository URL. Ownership must be transferred to `ZethetaIntern`. Must contain reproducible code and README.
Presentation (Optional)	Deck PDF (Max 10 slides) if a Report PDF was uploaded as the main Project Document.
Demo Video Link	Mandatory for PS4 (max 3 mins), optional for PS1-3. Include downloadable video URL on **Page 1 of Report** and top of **GitHub README**.

6. PRE-SUBMISSION VERIFICATION CHECKLIST

- [ ] **Selectable Text:** Verified that uploaded PDFs contain searchable text and are not image exports.
- [ ] **File Size Ceiling:** Ensured every uploaded document is under 20 MB.
- [ ] **GitHub Ownership:** Transferred repo ownership to `ZethetaIntern` and sent origin User ID + email to Zetheta.
- [ ] **Video Link Embed:** Verified video link is publicly accessible, downloadable, and listed on Page 1 of PDF.
- [ ] **Data & AI Citations:** Included comprehensive bibliography for all public data sources and AI tool disclosures.
- [ ] **Single Final Submission:** Confirmed submission completeness, as entries cannot be modified post-submission.

⚠️ **Hard Deadline Enforcement:** Submissions close automatically at the scheduled deadline displayed on the platform countdown. The submit button will lock instantly. Late entries will not be evaluated under any circumstances.